

TCX2005 SP1 2025 Exam

Your Name: _____

Your ID: _____

Question #: 1

TechSolutions Enterprise was planning a comprehensive IT infrastructure overhaul for their global operations. The CTO recognized that their current _____ problems, where the same customer data was stored inconsistently across multiple departmental systems, was creating significant operational inefficiencies and compliance risks.

For their new architecture, they needed to evaluate the _____ which analyzes both direct costs like hardware acquisition and indirect costs such as support and maintenance, since hardware and software typically account for only about 20% of this total measurement. The infrastructure team was debating whether to implement _____ computing to process data closer to its source for faster response times, or rely primarily on centralized cloud services.

To handle their diverse data requirements, they planned to implement a modern _____ that would serve multiple applications by centralizing data and controlling redundant information, while also deploying _____ databases to manage large volumes of both structured and unstructured data that traditional systems couldn't handle effectively.

For security, they recognized that the _____ Triad must be maintained, ensuring that data remains accessible to authorized users while protecting it from unauthorized access and maintaining its wholeness and accuracy. Given that people are often the weakest link in security, they planned to implement _____ principles where users would only have access to information required for their specific roles.

The company decided to adopt a _____ security model based on the principle of "never trust, always verify," requiring continuous verification regardless of user location. For their risk management strategy, they would apply safeguards to eliminate threats where possible, but for risks that were too expensive to mitigate, they would simply _____ them as part of their business operations. Finally, recognizing that security involves more than just technology, they emphasized that _____, process, and technology must all work together, with none being sufficient alone to ensure comprehensive protection.

- A. data redundancy and inconsistency, Total Cost of Ownership (TCO), edge, Database Management System (DBMS), non-relational (NoSQL), Confidentiality Integrity and Availability (C.I.A.), need-to-know, zero-trust, accept, people
- B. program-data dependence, Return on Investment (ROI), cloud, data warehouse, relational,

Authentication Authorization and Accounting (AAA), least-privilege, multi-factor, transfer, management

- C. data redundancy and inconsistency, Total Cost of Ownership (TCO), edge, business intelligence, distributed, Privacy Security and Compliance (PSC), role-based, defense-in-depth, mitigate, process
- D. system integration, infrastructure investment, quantum, business intelligence, distributed, Privacy Security and Compliance (PSC), role-based, defense-in-depth, mitigate, process
- E. data silos, operational expenditure, mobile, enterprise resource planning, centralized, Detect Respond and Recover (DRR), attribute-based, encryption-focused, eliminate, technology

Item Weight: 3.0

Question #: 2

TechManufacturing Corp is conducting a comprehensive risk assessment of their industrial IoT systems. They've identified that some risks are too expensive to mitigate compared to their potential impact, so they plan to _____ these minor risks. For their critical production systems, they're implementing firewalls and intrusion detection systems, which represents a _____ risk control strategy. However, for their legacy systems that are too vulnerable to secure effectively and no longer essential for operations, they've decided to _____ these systems entirely. The company recognizes that perfect information security is impossible and that security should be a balance between protection and _____.

- A. accept, defense, terminate, availability
- B. eliminate, mitigation, upgrade, efficiency
- C. ignore, acceptance, terminate, availability
- D. ignore, acceptance, modernize, convenience
- E. eliminate, prevention, isolate, productivity

Item Weight: 2.0

Question #: 3

SecureBank is implementing a comprehensive security overhaul across their global operations. They have 500 branches worldwide, handle millions of daily transactions, and must comply with various international banking regulations. Recent phishing attacks have compromised several employee accounts, and they

have discovered that 56% of their security incidents originate from internal sources.

Further, SecureBank discovered that their security framework needed to address three critical elements working together. The _____ element was their weakest link because employees were susceptible to social engineering and often bypassed security controls for convenience. To address this, they implemented a _____ security model where no one is trusted by default and continuous verification occurs regardless of user location. For their risk management strategy, they decided to purchase cyber insurance policies, which represents a _____ approach to risk control.

- A. people, zero-trust, transference
- B. people, zero-trust, mitigation
- C. process, least-privilege, mitigation
- D. network, encryption-based, acceptance
- E. technology, multi-factor, defense

Item Weight: 2.0

Question #: 4

Regional Healthcare Network operates 12 hospitals and 50 clinics across three countries. They are implementing an integrated patient records system that must comply with various privacy regulations while enabling real-time access to patient data for emergency situations. The system needs to handle medical imaging files, sensor data from patient monitoring devices, and traditional structured patient records.

Which combination of infrastructure components would best address their diverse data types and regulatory compliance requirements?

- A. Traditional relational databases with basic encryption and firewall protection
- B. Cloud-based data lake architecture with role-based access controls, audit logging, and data classification policies
- C. Separate systems for each country with manual data sharing processes
- D. Centralized mainframe system with terminal-based access for all locations
- E. All of the above

Item Weight: 2.0

Question #: 5

Global E-commerce Solutions was experiencing rapid growth and needed to redesign their data architecture. They discovered that their traditional _____ approach, where each department maintained separate customer files, was creating significant problems when generating integrated reports for management. To solve this, they implemented a modern _____ that would serve multiple applications by centralizing data and controlling redundant information. For their big data analytics needs involving customer reviews, social media mentions, and browsing patterns, they adopted _____ databases that could handle both structured and unstructured data more flexibly than traditional systems. Their business intelligence team wanted to use _____ to extract meaningful insights from customer feedback text, and _____ to analyze website navigation patterns and user engagement metrics.

- A. file processing, Database Management System (DBMS), non-relational (NoSQL), text mining, web mining
- B. data warehousing, data lake, relational, data mining, social media analytics
- C. file processing, Database Management System (DBMS), non-relational (NoSQL), data mining, sentiment analysis
- D. system integration, enterprise resource planning, distributed, sentiment analysis, clickstream analysis
- E. database management, business intelligence, cloud-based, natural language processing, conversion optimization

Item Weight: 2.0

Question #: 6

Regional Bank Network was upgrading their core banking infrastructure to support digital transformation. Their IT director knew that to handle customer transaction data effectively, they needed to address the _____ problem where the same customer information was stored inconsistently across multiple departmental systems.

For their new architecture, they decided to implement _____ which would store data in computer main memory rather than on disk storage to achieve faster transaction processing. To support their mobile banking expansion, they needed to consider _____ security challenges, particularly the risk of _____ where attackers could drive around bank branches attempting to detect and breach wireless access points.

Finally, for their business intelligence needs, they planned to use _____ to

analyze customer behavior across multiple dimensions such as account type, geographic region, and time periods.

- A. program-data dependence, cloud computing, network, phishing attacks, data mining
- B. data redundancy and inconsistency, in-memory computing, wireless, war driving, Online Analytical Processing (OLAP)
- C. data redundancy and inconsistency, edge computing, mobile, war driving, text mining
- D. system integration, quantum computing, internet, denial-of-service, Online Analytical Processing (OLAP)
- E. system integration, quantum computing, internet, war driving, Online Analytical Processing (OLAP)

Item Weight: 2.0

Question #: 7

National Healthcare Systems was implementing a comprehensive electronic health records platform. Their CIO recognized that for sensitive patient data, the team needed to address _____ concerns through proper controls, while their infrastructure decisions needed to consider the _____ beyond just purchase price.

After evaluating available options, they decided to use a _____ integration approach to connect their various systems rather than point-to-point connections. For the hardware implementation, they considered the growing trend of _____ to process data closer to its source for faster response times in emergency situations.

To ensure proper governance, they adopted a _____ model where all IT decisions were made by a central department rather than individual hospital units making independent technology choices.

- A. intellectual property, Total Cost of Ownership, hub-and-spoke, edge computing, centralized
- B. system quality, procurement costs, service bus, cloud computing, federated
- C. information rights and privacy, operational efficiency, API-based, quantum computing, outsourced
- D. ethical and social issues, hardware inventory, blockchain-based, green computing, decentralized
- E. intellectual property, Total Cost of Ownership, blockchain-based, green computing, decentralized

Item Weight: 2.0

Question #: 8

What percentage of the Total Cost of Ownership (TCO) is typically accounted for by hardware and software?

- A. About 80%
- B. About 50%
- C. About 20%
- D. About 10%
- E. About 5%

Item Weight: 2.0

Question #: 9

Which of the following is NOT a typical outcome of implementing advanced IT infrastructure in a ride-hailing service?

- A. Enhanced service reliability
- B. Expanded financial inclusion
- C. Regional market leadership
- D. Reduced customer base
- E. Increased market competition

Item Weight: 2.0

Question #: 10

HealthTech Singapore, a major healthcare provider, embarked on implementing a new patient management system. First, they analysed where they could differentiate themselves from competitors according to the _____ framework. To maintain competitive viability in an industry where innovations are quickly copied, they needed to address the _____ challenge. After evaluating their operations, they determined which processes were core to their success using the _____ model to map out activities where IS investments would create the most value. For patient data handling, they needed to address _____ concerns through comprehensive policies, while ensuring their implementation recognized both the technological and cultural changes required by following a _____ approach to change management.

- A. Porter's competitive forces, competitive advantage paradox, Value Chain, information rights and privacy, socio-technical
- B. PESTLE analysis, business-IT alignment, Resource-Based View, system quality, transformational leadership
- C. Porter's competitive forces, competitive advantage paradox, Business Process Reengineering, intellectual property, organizational development
- D. Strategic Grid, vendor lock-in, Enterprise Systems, data governance, technical-rational
- E. Strategic Grid, vendor lock-in, Enterprise Systems, system quality, transformational leadership

Item Weight: 2.0

Question #: 11

Which of the following is NOT identified as a key technology trend that raises ethical issues?

- A. Computing power doubling approximately every 18 months
- B. Data analysis advances enabling profiling
- C. Reduced cost of long-term data storage
- D. Increased reliance on open-source software

Item Weight: 2.0

Question #: 12

A global logistics firm is undertaking a multi-year digital transformation initiative that involves transitioning from legacy mainframe systems to a cloud-based, AI-driven logistics platform. The new system promises predictive routing, real-time tracking, and cost optimization. However, during the rollout, regional offices continue to rely on legacy systems for daily operations, leading to data inconsistencies, coordination issues, and employee resistance. The CIO insists on a phased implementation strategy to minimize disruption. Which of the following best illustrates the core challenge of managing strategic transitions in this scenario?

- A. Aligning the new system's user interface with employee expectations
- B. Choosing between cloud vendors based on cost and scalability

- C. Ensuring continuity of business operations while transforming core processes and systems
- D. Maintaining cybersecurity policies during the migration process
- E. Choosing between cloud vendors based on vendor success stories and review scores

Item Weight: 2.0

Question #: 13

SS Airlines sought to improve its customer experience and operational efficiency through digital transformation. First, they identified how their ticketing and customer service processes created value for customers, following the _____ model. To counter the threat of travel booking platforms with significant _____ over suppliers, they determined that enhancing customer loyalty through personalized service would be their strategic advantage. Rather than developing all components in-house, they adopted a hybrid _____ approach, purchasing standard reservation systems while developing custom analytics for their loyalty program. Before implementation, they conducted a thorough analysis of _____ factors, particularly focusing on how data localization requirements in different jurisdictions would affect their global system architecture.

- A. Business Value Chain, bargaining power, build vs. buy, political
- B. Organizational-Environmental, network effects, software acquisition, regulatory
- C. Business Value Chain, network effects, software acquisition, political
- D. Enterprise Application, customer intimacy, outsourcing, technological
- E. Business Value Chain, market niche control, in-house development, social

Item Weight: 2.0

Question #: 14

A major entertainment retailer dominated the video rental market throughout the 1990s and early 2000s with over 9,000 physical stores worldwide. When a small startup began offering DVD-by-mail subscriptions in 1997, the retailer's executives dismissed the model as niche. In the mid-2000s, the startup began shifting toward streaming video content online. The retail giant had the opportunity to purchase this startup for \$50 million in 2000 but declined. By 2010, the retailer was losing market share rapidly as consumers embraced digital streaming. Despite a late attempt to launch their own streaming service and eliminate late fees, the company filed for bankruptcy in 2010. By 2014, all but a single store had closed.

Which digital adaptation strategy BEST describes what this company implemented?

- A. Fight back - They aggressively competed by creating their own digital platform and acquiring smaller streaming competitors
- B. Double down - They focused on improving their core business by enhancing the in-store experience and exclusive partnerships
- C. Retrench - They consolidated stores and sought regulatory protection against digital competitors
- D. Failing to adapt - They recognized the threat too late and made insufficient changes to their business model to survive digital disruption
- E. All of the above

Item Weight: 2.0

Question #: 15

Given: Sample Report on- Regional Sales Performance Q4 2024

Product Line	Region	Actual Sales (\$)	Planned Sales (\$)	Actual vs Planned
Electronics	North	825,000	800,000	1.03
	South	675,000	700,000	0.96
	East	945,000	900,000	1.05
	West	555,000	600,000	0.93
Total Electronics		3,000,000	3,000,000	1.00
Appliances	North	445,000	400,000	1.11
	South	385,000	400,000	0.96
	East	515,000	500,000	1.03
	West	355,000	400,000	0.89
Total Appliances		1,700,000	1,700,000	1.00

What is the overall performance ratio for the West region across all product lines?

- A. 0.91
- B. 0.92
- C. 0.93
- D. 0.89
- E. 0.88

Item Weight: 2.0

Question #: 16

According to the Decision Framework for Acquisition, which of the following tasks would MOST likely be classified as a "Critical Task" rather than a "Strategic Task"?

- A. Trading algorithms for a financial firm
- B. Customer relationship management for a luxury brand
- C. Payroll system for a manufacturing company
- D. Proprietary machine learning algorithm for product recommendations

Item Weight: 2.0

Question #: 17

In the Build vs Buy decision framework, which scenario would BEST justify a "build" decision?

- A. The organization needs a standard email system with minimal customization
- B. The timeline for implementation is extremely short and resources are limited
- C. The system supports unique business processes that create competitive advantage
- D. Several proven market solutions exist that meet 90% of requirements
- E. All of the above

Item Weight: 2.0

Question #: 18

A retail company is experiencing declining profit margins despite steady sales. Senior management suspects inefficiencies in internal operations and wants to

identify specific areas, such as inbound logistics, operations, marketing, and after-sales service, where cost savings or value enhancements can be made. Which analytical framework should they use to systematically evaluate and optimize each stage of their internal activities for competitive advantage?

- A. Porter's Five Forces Analysis
- B. Value Chain Model
- C. PESTLE analysis
- D. Balanced Scorecard
- E. Network economics

Item Weight: 2.0

Question #: 19

A budget airline launches a mobile app that offers exclusive low fares, in-app check-ins, and personalized travel packages. The airline also uses dynamic pricing algorithms based on customer profiles.

Which competitive force and IS strategy pair is most relevant here?

- A. Threat of substitutes –Process automation
- B. Bargaining power of buyers –Personalization and lock-in
- C. Rivalry among competitors –Cost leadership
- D. Threat of new entrants –Market penetration via mobile strategy
- E. Bargaining power of buyers– Cost leadership

Item Weight: 2.0

Question #: 20

BrightTech Ltd. is a mid-sized electronics manufacturer with four core departments: Procurement, Production, Sales, and Finance. Each department currently relies on its own stand-alone software: ProcurePro for purchasing, ShopFloor™ for shop-floor scheduling, SalesPlus for order entry, and FinTrack for accounting.

Because these systems do not 'talk' to one another, purchase orders have to be re-entered for production, sales forecasts rarely align with stock levels, and invoicing is often delayed by manual data reconciliation. Last quarter alone, these inefficiencies added an extra 10 days to order fulfillment and generated multiple

billing disputes with key customers.

To eliminate duplication, reduce delays, and improve cross-departmental visibility, BrightTech's executive team is evaluating the deployment of a single enterprise application that would unify all critical business functions.

Drawing on the scenario above, clearly and concisely explain **how** and **why** BrightTech Ltd. might implement an enterprise application.

What steps could be involved in this implementation process?

Case Study - HealthBridge: When the Weakest Link Is Human

HealthBridge is a mid-sized private healthcare group operating seven clinics across a major metropolitan area. Like many healthcare organisations, it holds highly sensitive patient data including medical histories, prescription records, and insurance details across an integrated information system. In 2023, HealthBridge undertook a major digital transformation, migrating much of its patient data to a cloud-based Electronic Health Record (EHR) platform managed by a third-party vendor, CareVault Technologies.

In March 2024, HealthBridge discovered that approximately 42,000 patient records had been exfiltrated over a six-week period. Investigations revealed a multi-stage attack. A staff member in the finance department had received an email appearing to come from CareVault's support team, warning of an urgent account verification requirement. The employee clicked a link in the email, entered their login credentials on a convincing but fraudulent website, and unknowingly handed attackers access to the corporate network. This technique, known as phishing, exploited the trust the employee placed in what appeared to be a legitimate communication.

Once inside the network, the attackers moved laterally, eventually accessing the EHR platform. The breach was possible partly because HealthBridge had not enforced multi-factor authentication (MFA) on its EHR system, and the compromised staff account had been granted access rights well beyond those needed for a finance role. The principle of least privilege had not been applied consistently across the organisation.

The consequences were significant. HealthBridge faced regulatory scrutiny from the national data protection authority, reputational damage, and the cost of notifying affected patients. Post-incident analysis also revealed that CareVault's service level agreement had not clearly defined incident response responsibilities, leaving both parties uncertain about who was obligated to act and when. HealthBridge's incident response plan, which had been drafted two years earlier, had never been tested or updated to account for cloud-hosted systems.

In the aftermath, HealthBridge's leadership commissioned a security overhaul. This included enforcing MFA across all systems, adopting a zero-trust architecture that required continuous verification regardless of whether users were inside or outside the corporate network, redefining access rights according to need-to-know and least privilege principles, conducting regular phishing simulation training for all staff, and renegotiating vendor contracts to include clear security responsibilities and incident response obligations. The chief information officer noted that the breach was ultimately a failure not of technology alone, but of people, processes, and governance working in combination.

Question #: 21

The initial attack on HealthBridge involved a staff member being deceived into entering credentials on a fraudulent website that mimicked a trusted vendor. This technique is best described as:

- A. War driving, because the attacker exploited wireless network vulnerabilities to gain access.
- B. Phishing, because the attacker used a deceptive communication to trick the user into revealing credentials.
- C. A denial-of-service attack, because it disrupted access to the EHR platform.
- D. SQL injection, because the attacker exploited a software programming error to access the database.

Item Weight: 2.0

Question #: 22

After the breach, HealthBridge renegotiated its contract with CareVault to clearly define security responsibilities and incident response obligations. Which risk management strategy does this action best represent?

- A. Termination, because HealthBridge removed a vulnerable system from its operating environment.
- B. Acceptance, because HealthBridge acknowledged the residual risk and chose to continue using CareVault.
- C. Transference, because HealthBridge shifted a portion of the security risk and responsibility to a third party through contractual terms.
- D. Mitigation, because HealthBridge reduced the likelihood of a breach occurring by installing additional technical controls.

Item Weight: 3.0

Question #: 23

The HealthBridge breach resulted in 42,000 patient records being exfiltrated without authorisation. In terms of the C.I.A. Triad, which dimension of information security was most directly violated?

- A. Confidentiality, because patient records were accessed by unauthorised parties.
- B. Integrity, because the attacker corrupted or altered patient data after gaining access.
- C. Availability, because the EHR system was taken offline and staff could no longer access records.
- D. Accountability, because HealthBridge was unable to identify which staff member was responsible for the breach.

Item Weight: 3.0

Question #: 24

The CIO of HealthBridge concluded that the breach was a failure of people, processes, and technology working in combination. Using evidence from the case, explain how each of the three elements contributed to the breach.

Item Weight: 6.0

Question #: 25

HealthBridge adopted a zero-trust architecture as part of its post-breach security overhaul.

Explain what zero-trust means and why it would have been particularly relevant to preventing or limiting the damage from this breach.

Item Weight: 6.0

END
